PHYS 6031
Spring 2024
Homework 4
Due: Friday 16 February 2024 at 4:00 pm in my Clippinger mailbox

1. Consider the electrostatic potential Φ in a semi-infinite strip in the x-y plane covering the region $0 \leq x \leq a$, $y \geq 0$, with boundary conditions

$$\begin{aligned} \Phi(x, 0) &= V_0 \quad \text{(constant)} \\ \Phi(0, y) &= 0 \\ \Phi(a, y) &= 0 \\ \Phi(x, y \to \infty) &= 0 \end{aligned}$$

Assume that Φ satisfies the 2-D Laplace Equation.

(a) Find Φ within the strip using separation of variables.

(b) Show that the infinite series for Φ can be summed by realizing that it is the imaginary part of the derivative of a geometric series. What is this result for Φ?

(c) Show that Φ can be written as the real or imaginary part of an analytic function.

2. Consider a semi-infinite line charge located on the $+z$ axis, with a charge per unit length given by:

$$\lambda(z) = \begin{cases} \lambda_0 \exp(-z/a) & z \geq 0 \\ 0 & z < 0 \end{cases},$$

where λ_0 and $a > 0$ are constants. Using spherical coordinates, find the electrostatic potential everywhere, assuming $\Phi(r \to \infty) = 0$. It is sufficient to express you answer in terms of definite integrals over r.

3. Suppose electrostatic potential Φ is given on the surface of a sphere of radius a:

$$\Phi(a, \cos\theta) = V_0(1 + \cos\theta)$$

The is also a point charge of strength q located at $(x, y, z) = (0, 0, a/2)$. Otherwise, the space inside the sphere is empty.

(a) Determine Φ inside the sphere by assuming $\Phi = \Phi_q + \Phi_0$, where Φ_q is the potential due to the point charge:

$$\Phi_q = \frac{q}{4\pi\epsilon_0 |\vec{\boldsymbol{x}} - a\hat{\boldsymbol{z}}/2|}.$$

Note that $\nabla^2\Phi_0 = 0$ and Φ_0 should have no singularities inside the sphere.

(b) Show that the same result can be obtained using a Green Function. The Green Function for the volume inside a sphere is given by Jackson, Eq. (3.125), with $a \to 0$. You may use this equation as your starting point.

4. Determine the electric field everywhere outside of an uncharged conducting sphere of radius a that is placed in a constant external field of magnitude $E_0\hat{z}$.